

Fundamentals of Meteorological Observations

Unit 1: Surface Observatory Network & Siting Standards

Trainer notes — for internal training use on the Capacity Connect platform.

Learning Objectives

- Describe the structure of IMD's surface observatory network
- Apply WMO/IMD siting standards when assessing an observation enclosure
- Explain the standard synoptic observation schedule

IMD's Observatory Network

IMD maintains a nationwide network of surface observatories that includes full-fledged departmental observatories, part-time observatories operated jointly with state governments, airports and other agencies, and a rapidly growing layer of Automatic Weather Stations (AWS) and Automatic Rain Gauges (ARG). Observatories are graded by the instruments they carry and the observations they are required to take - a Class I station typically records surface parameters plus upper-air soundings, while smaller stations may report only surface parameters at fixed hours.

Siting Standards

Instrument enclosures follow WMO and IMD siting guidelines to keep readings representative and comparable across stations: level, grass-covered ground; a minimum clearance from buildings, trees and paved surfaces (generally at least twice the height of the nearest obstruction); a standard thermometer screen height of about 1.25 m; and a location clear of local heat sources such as exhaust vents or concrete aprons. Poor siting is one of the most common causes of unrepresentative temperature and wind data.

Observation Schedule

Surface synoptic observations are taken at fixed hours coordinated in UTC (0000, 0300, 0600, 0900, 1200, 1500, 1800, 2100), with busier stations, airports and cyclone-watch stations reporting hourly or half-hourly. This synchronised schedule is what allows IMD, and international partners exchanging data through the WMO Global Telecommunication System, to build a consistent snapshot of the atmosphere across the country at the same instant.

Summary

A representative observation depends as much on correct siting and a disciplined schedule as it does on the instrument itself - this unit is the foundation for everything that follows in the course.